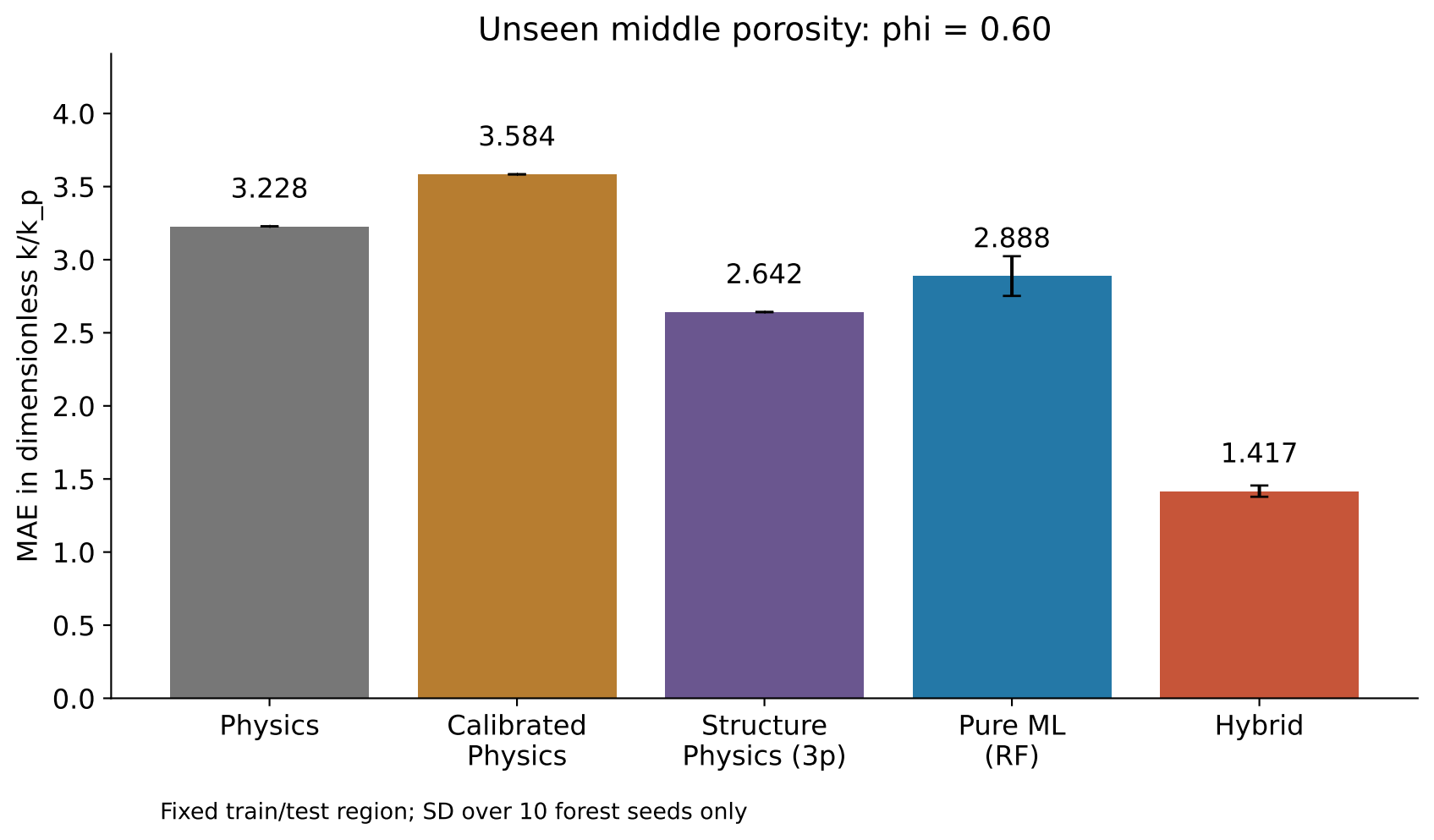
Unseen middle porosity: phi = 0.60
MAE in dimensionless k/k_p
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
3.228
3.584
2.642
2.888
1.417
Physics
Calibrated Physics
Structure Physics (3p)
Pure ML (RF)
Hybrid
Fixed train/test region; SD over 10 forest seeds only